Current Setup & Baselines

Policy, transfer function, and the four training baselines run 2026-10-06/07

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> **Note:** Quick reference for picking this back up in the lab. Full detail lives in `docs/rl-training-setup.typ` (policy/TF/reward) and `docs/experiments/2026-10-07-search-distilled-pretraining-results.md` (the baseline runs below, with findings and caveats).

1. Policy and transfer function

Current checkpoint family: **`oneshot-v9`**, `--action-mode residual`, `--reward-mode target`. The transfer function is **label-scoped**: a voxel segmented as class c is shaded only by c's own peak, so the eight goal classes (skeleton, lungs, heart, vessels, liver, kidneys, spleen, soft) no longer fight over the same HU range. Observation size 97, action size 24.

2. Experiment design

2.1. The question

Does a second kind of imitation data (search-distilled examples) actually make the policy better – and if so, why? Everything below exists only to answer that one question cleanly.

2.2. One variable at a time

Running only the final recipe and reporting a good number would prove nothing: if five things changed at once and the score went up, there is no way to tell which of the five did it, or whether it was noise. The discipline instead is to change **exactly one ingredient** per step, freeze everything else, and measure. Any difference observed is then attributable to that one ingredient, because nothing else moved.

2.3. The staircase

Step	Isolates
E1 → E2	Does hindsight pretraining help at all, over nothing?
E2 → E2+search	What does search-distilled data add, on top of hindsight (unchanged)?
E2+search → E3	Does continuous brightness anchoring fix what the previous step exposed, without undoing its gains?

E3 was not planned from day one – it exists only because the E1/E2/E2+search comparison surfaced a problem worth fixing. That is the point of running the staircase instead of jumping straight to a guessed-at final recipe.

2.4. The brightness blind spot, and why per-kind scoring caught it

A single blended score would have made E2 look like a clean win over E1. Breaking the score down **per instruction kind** (absolute, relative, brightness, …) is what exposed that E2 was dramatically worse at brightness specifically than even doing nothing – invisible in the average, obvious in the breakdown.

> **Why:** None of the 4000 hindsight demonstrations are brightness-type, so the actor is fit hard to four topics and never once shown the fifth – and nothing during RL training brings brightness back in, since hindsight-generated training episodes are the same four topics. Like cramming four of five exam topics: the four get very good, and the fifth quietly atrophies rather than staying neutral. E3's fix is to

> keep a trickle of brightness examples flowing throughout training, the same continuous reinforcement the other four topics already get.

2.5. Why three seeds

SAC's outcome is not deterministic – a different random seed can land noticeably differently even with the exact same recipe, so one good run could just be a lucky draw. The fix is this project's own standing bar: rerun the winning recipe on three seeds and require the **median** result (not any single run) to clear median attainment > 0 with share improved $> 50\%$. E3 was only called a result once seeds 1 and 2 agreed with seed 0, not after seed 0 alone.

3. Training baselines

All four runs: 150k timesteps, `--hindsight-ratio 0.25 --balance-classes`, seed 0, 200-episode validation eval every 10k steps. Only the pretrain source differs – this isolates what each teacher signal contributes.

Run	Pretrain source	Out dir
E1 (control)	none	`e1_v9_150k_seed0_rewardfix`
E2	hindsight only, 4000 ep	`e2_v9_bc_150k_seed0_rewardfix`
E2+search	hindsight (4000) + search-distilled (1000 @ 200 evals)	`e1_v9_150k_seed0_searchmix`
E3	hindsight+brightness (4000, `--hindsight-brightness 0.1`) + search-distilled (1000)	`e3_v9_searchmix_bright_150k_seed0`

Search-distilled pretraining (`--search-pretrain-data`) is the new teacher this session added: `hill_climb` solutions on real sampled instructions, distilled into the actor alongside the existing hindsight demonstrations (`rl.pretrain.collect_search`, `OneShotEnv.search_policy_action`).

3.1. Results (mean of the last 3 validation evals, steps 130k/140k/150k)

Metric	E1	E2	E2+search	E3
median attainment	−0.034	+0.088	+0.109	**+0.179**
absolute	−0.096	**+0.193**	+0.109	+0.176
relative	−0.474	**−0.052**	−0.208	−0.068
compound	−0.047	+0.047	+0.001	**+0.088**
show only	0.054	0.070	+0.076	**+0.096**
brightness	−1.278	−5.204	−2.379	**+0.246**

> **Headline finding:** Hindsight-only pretraining (E2) causes a severe brightness-instruction regression (-5.204) – none of its 4000 demonstrations are brightness-type, and nothing during RL training ever anchors that behaviour. Search-distilled data partially repairs it (E2+search: −2.379); adding continuous brightness hindsight during RL training (E3) fixes it outright (+0.246) while also topping every other metric.

3.2. E3, confirmed across 3 seeds

Metric	seed 0	seed 1	seed 2	median-of-3
median attainment	+0.179	+0.143	+0.153	**+0.153**
brightness	+0.246	+0.219	−0.010	**+0.219**

Metric	seed 0	seed 1	seed 2	median-of-3
share improved	0.718	0.708	0.677	0.708

Clears this project's own decision gate (median > 0 and share improved $> 50\%$ on all three seeds). Brightness is strongly positive on 2 of 3 seeds and near-neutral (not negative) on the third – a real improvement, not yet a universal per-seed win on that one instruction kind.

> **Note:** Not yet done: a held-out **test**-split evaluation (`rl.vis_eval + tools.per_class_eval`) – everything above is validation-split (`eval_progress.csv`) only.

4. Viewer / VR status

The app (`server.py`) currently points at `out/rl_v4/e3_v9_searchmix_bright_150k_seed0/best.zip` via the new `POLICY_PATH` env-var override (`policy.py`). Networking to a Vive Focus 3 over the LAN works once bound to `--host 0.0.0.0` with the `certs/lan.pem` TLS cert (needed for WebXR/mic). Open: VR framerate is single-digit on the Focus 3's Snapdragon XR2 – a debug line (`xr=/auto=/sd=/shade=/dim=`) was added to the in-VR HUD to get real numbers next session, plus an `?xrscale=` URL override to test resolution-bound-ness. To resume next time in the lab.